

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 27

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Any Reminders? Any Questions?

- Teaching evaluations are open!
- Exam 3: Friday May 1 – Ch 13.1-13.3 and 16.1-16.8
- Review for Exam 3 during class on Thursday!
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104

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ALVARO: Start the recording!



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Multi-Variable Calculus

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Review of Vector Calculus

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Questions?

1. Vector Functions and Curves

- P1.** Find the unit tangent vector $\vec{T}(t)$ for $\vec{r}(t) = \langle t, \frac{2}{3}t^{3/2}, t^2 \rangle$ at $t = 1$.
- P2.** Find the length of the curve $\vec{r}(t) = \langle e^t \cos t, e^t \sin t, e^t \rangle$ for $0 \leq t \leq \ln(2)$.
- P3.** Consider the curve C formed by the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $z = 1 - y$.
- (a) Find a parametrization $\vec{r}(t)$ for C .
 - (b) Find $\vec{r}'(t)$ and the line integral $\oint_C \langle -y, x, z \rangle \cdot d\vec{r}$.

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2. Conservative Fields and Green's Theorem

- P4.** Use Green's Theorem to evaluate $\oint_C (xy + e^{x^2}) dx + (x^2 + \sin(y^2)) dy$ where C is the boundary of the rectangle with vertices $(0, 0)$, $(3, 0)$, $(3, 2)$, and $(0, 2)$.
- P5.** Show that $\vec{F} = \langle y \cos(xy), x \cos(xy) + 2y \rangle$ is conservative and find a potential function f . Use it to evaluate $\int_C \vec{F} \cdot d\vec{r}$ from $(0, 0)$ to $(1, \pi)$.

3. Parametric Surfaces and Surface Area

- P6.** Let S be the part of the plane $3x + 2y + z = 6$ in the first octant.
- (a) Find a parametrization $\vec{r}(x, y)$ for S .
 - (b) Express the domain D as a **Type I (Type- x)** domain.
 - (c) Compute $\vec{r}_x \times \vec{r}_y$ and find the surface area of S using integration.
- P7.** Find the area of the surface $z = xy$ that lies within the cylinder $x^2 + y^2 = 4$.

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4. Flux and Vector Integrals

P8. Let S be the part of the cone $z = \sqrt{x^2 + y^2}$ between $z = 0$ and $z = 2$.

- (a) Write a parametrization $\vec{r}(r, \theta)$ for S using **polar coordinates**.
- (b) Find the normal vector $\vec{r}_r \times \vec{r}_\theta$.
- (c) Evaluate the flux of $\vec{F} = \langle x, y, z^4 \rangle$ through S with upward orientation.

5. Stokes' and Divergence Theorems

P9. Let $\vec{F} = \langle z^2, x^2, y^2 \rangle$. Use Stokes' Theorem to evaluate $\oint_C \vec{F} \cdot d\vec{r}$, where C is the triangle with vertices $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$, oriented counterclockwise when viewed from above.

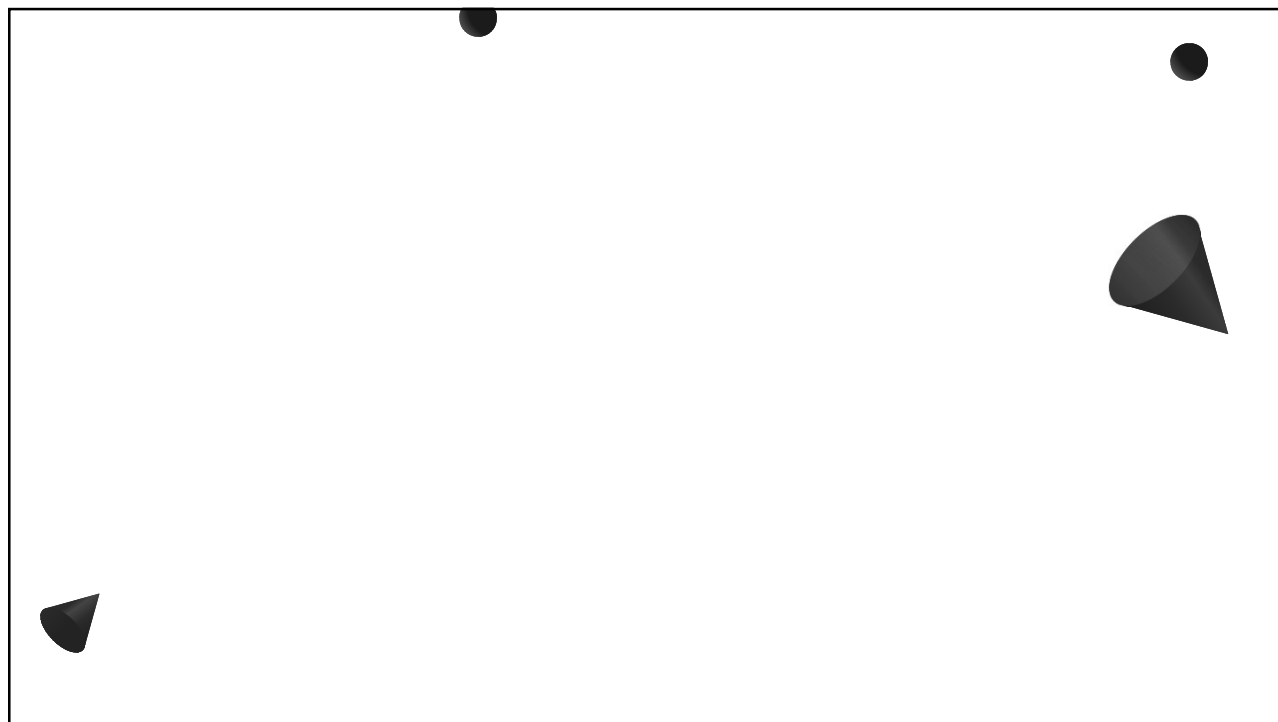


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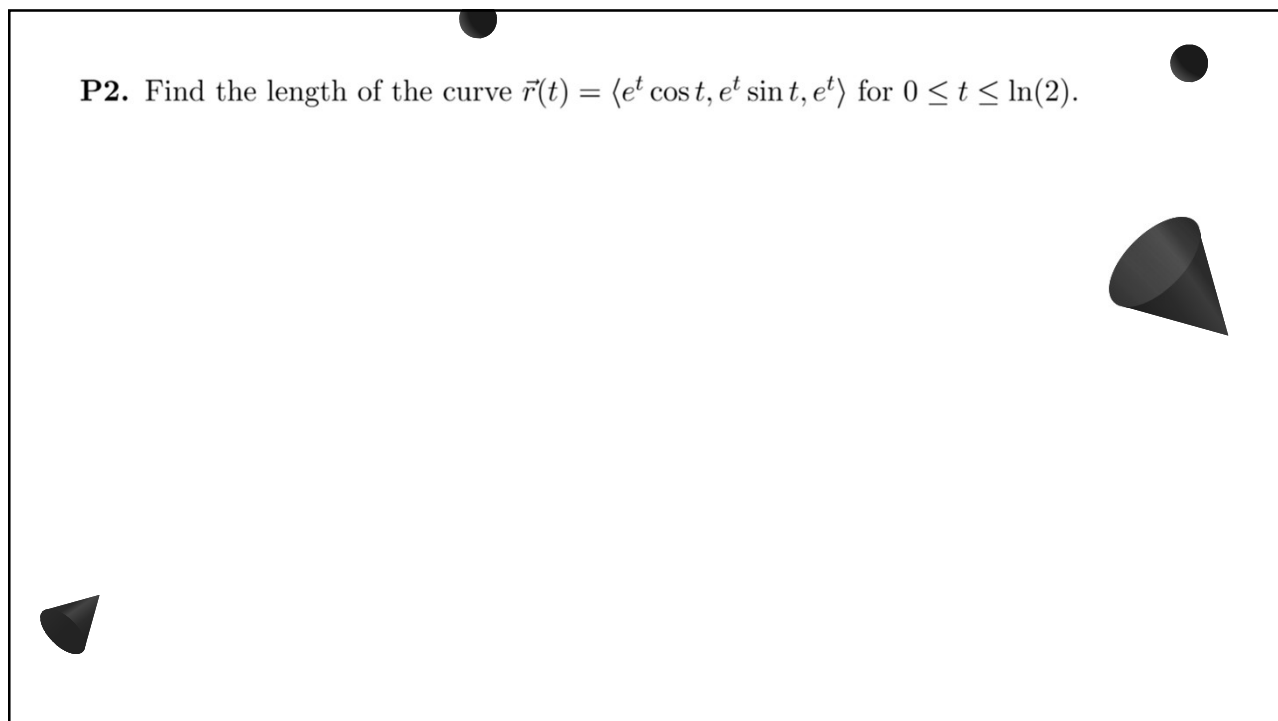
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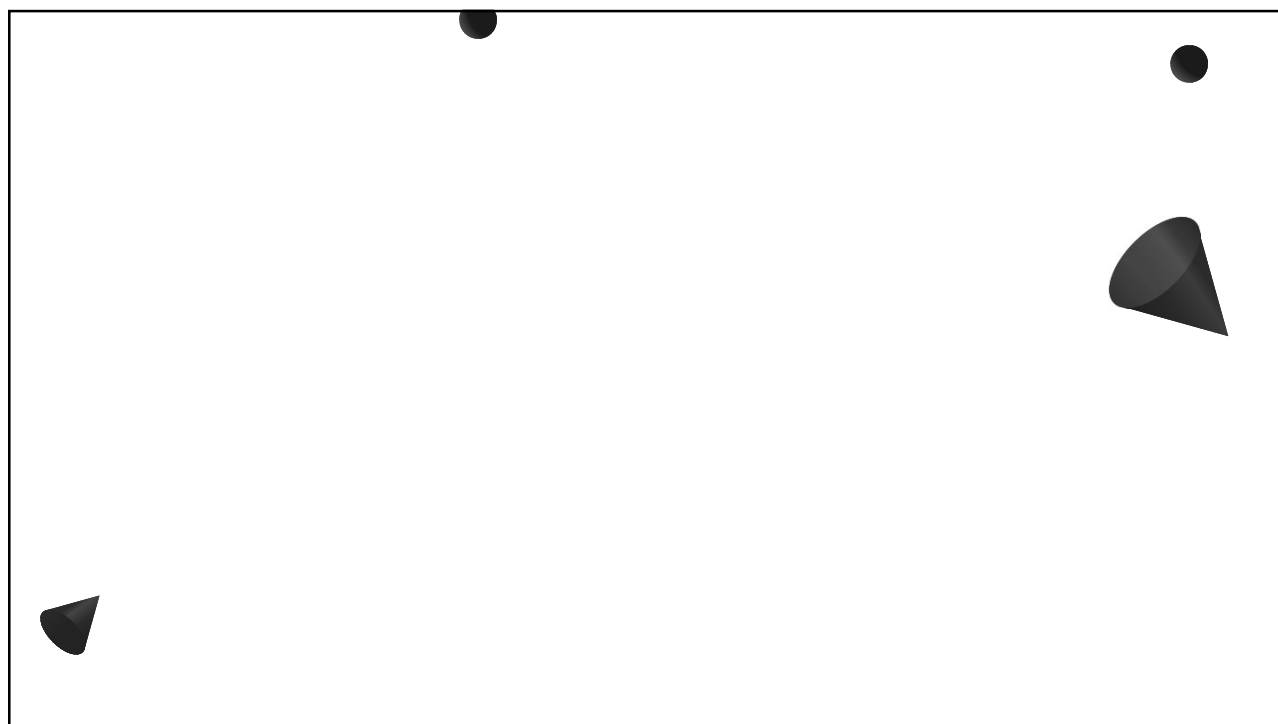


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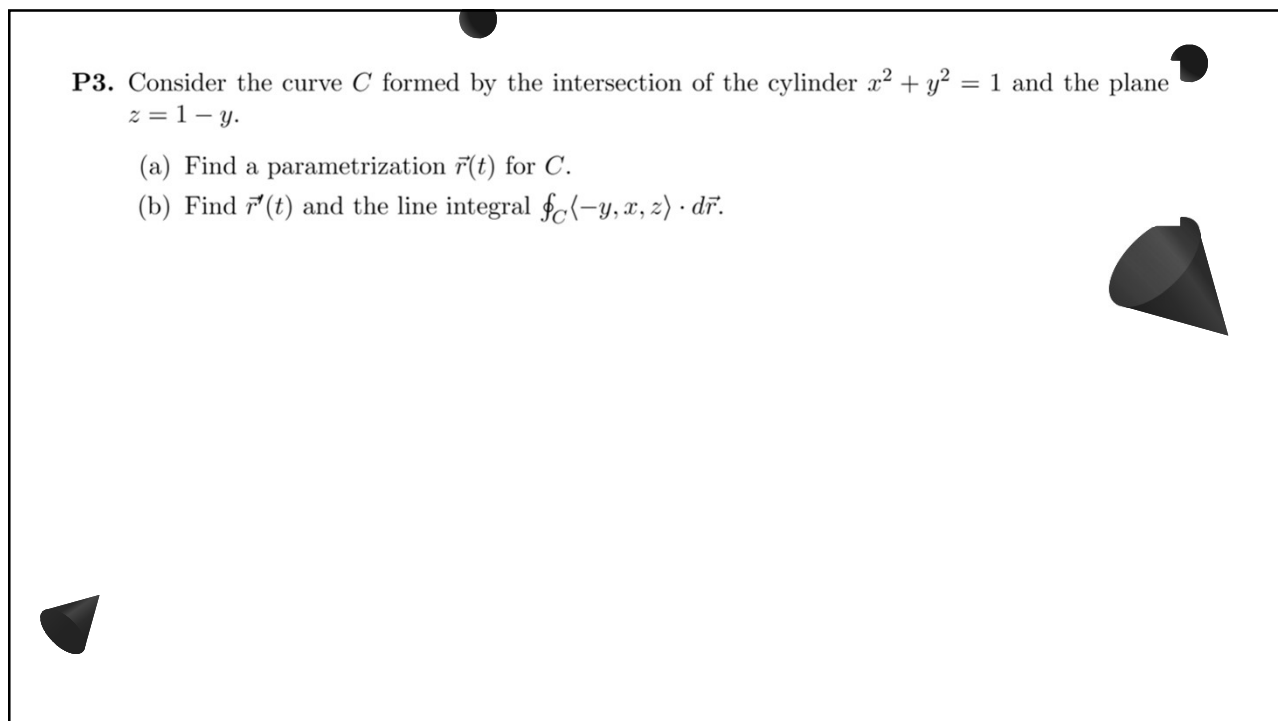


P2. Find the length of the curve $\vec{r}(t) = \langle e^t \cos t, e^t \sin t, e^t \rangle$ for $0 \leq t \leq \ln(2)$.

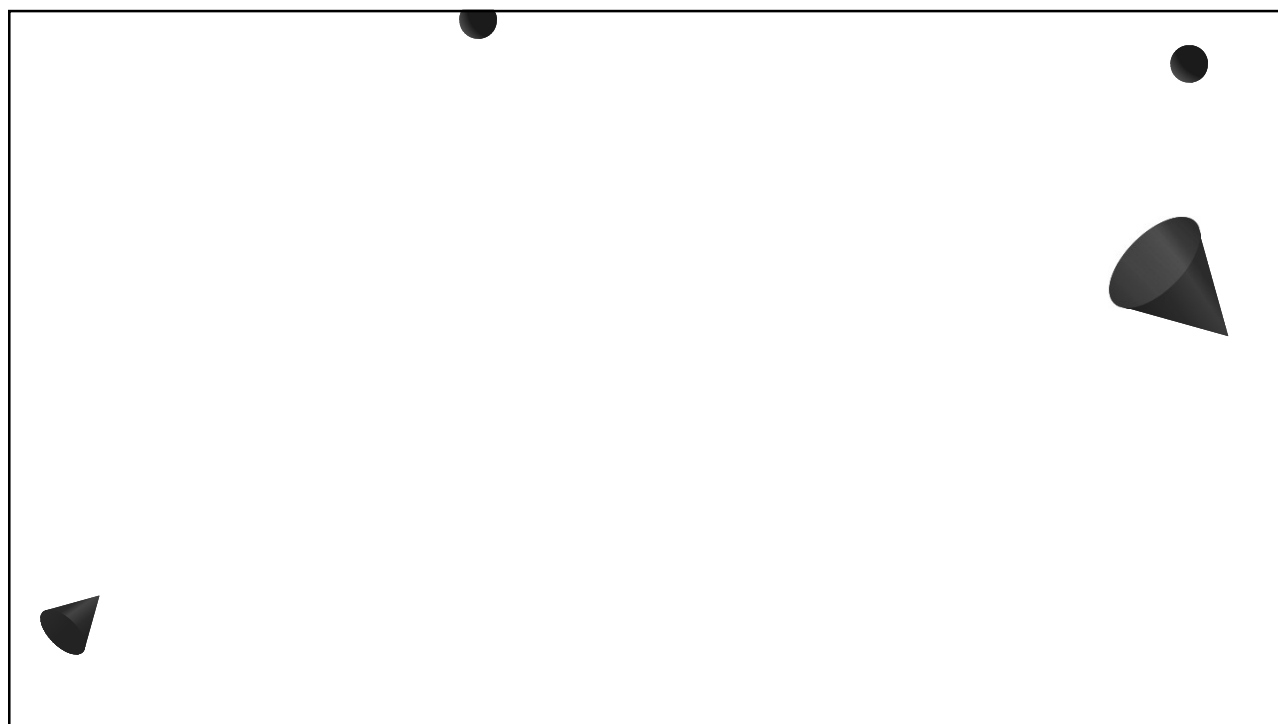
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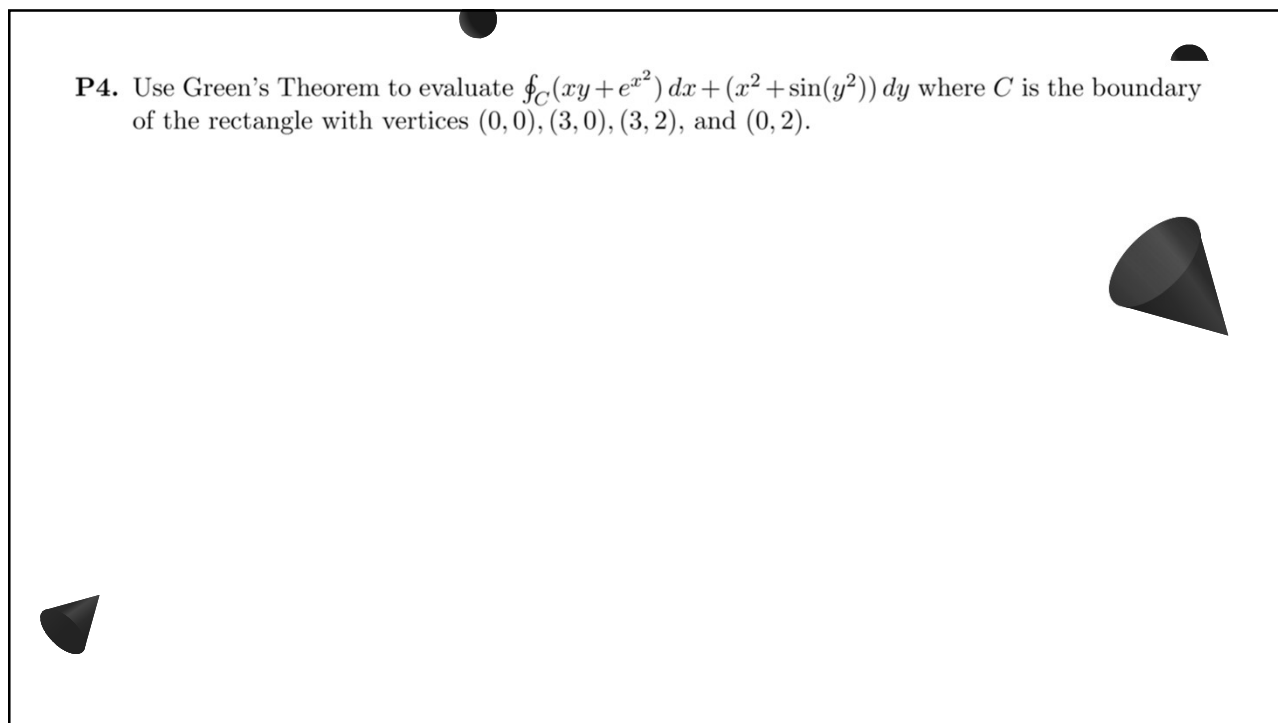
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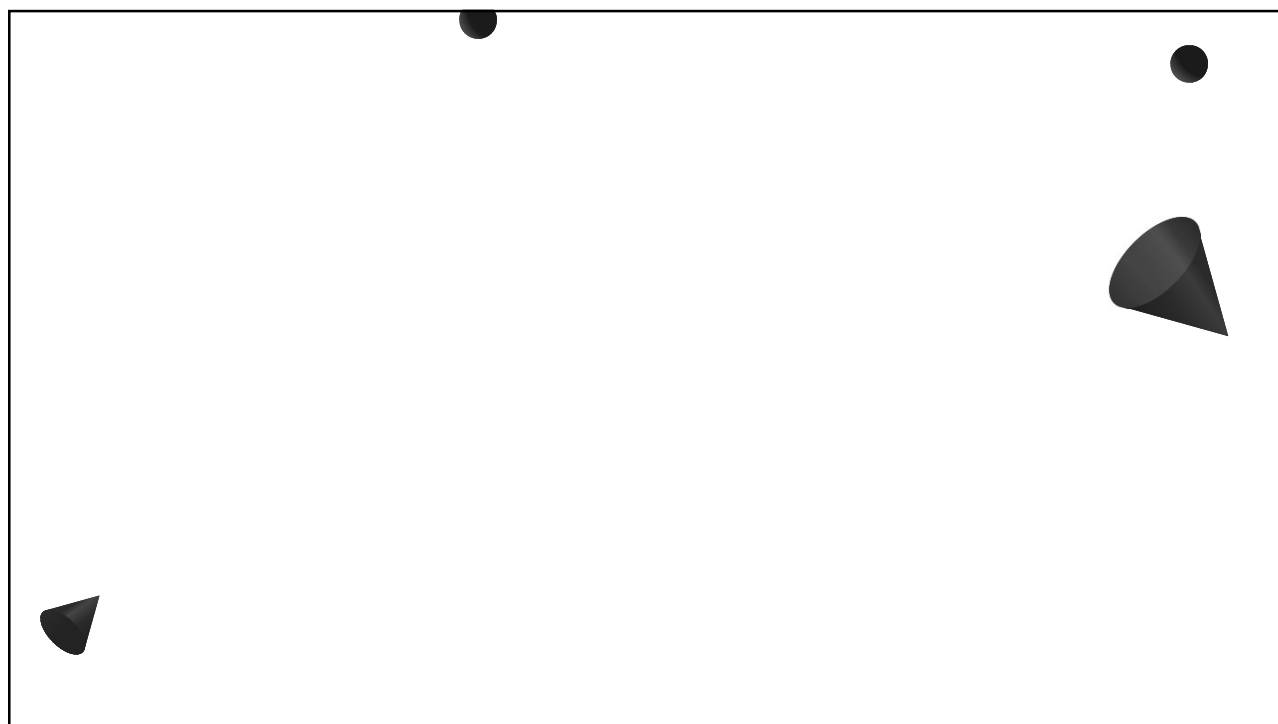
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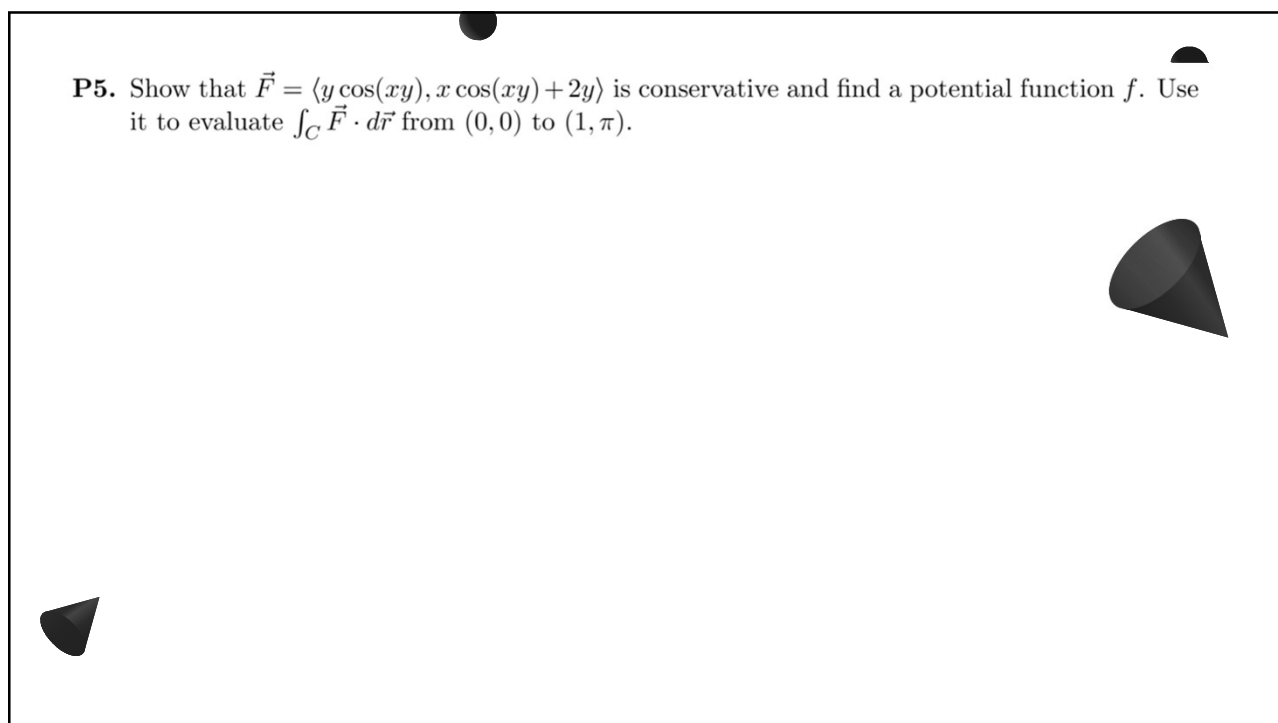
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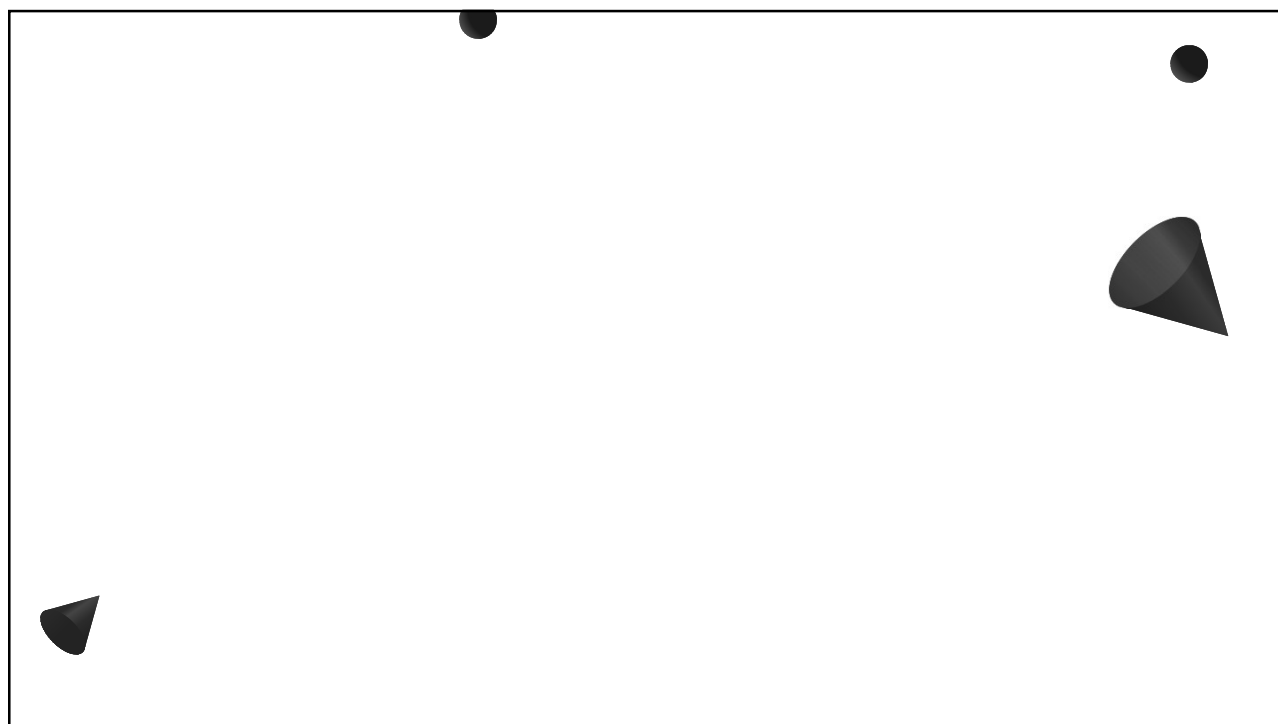
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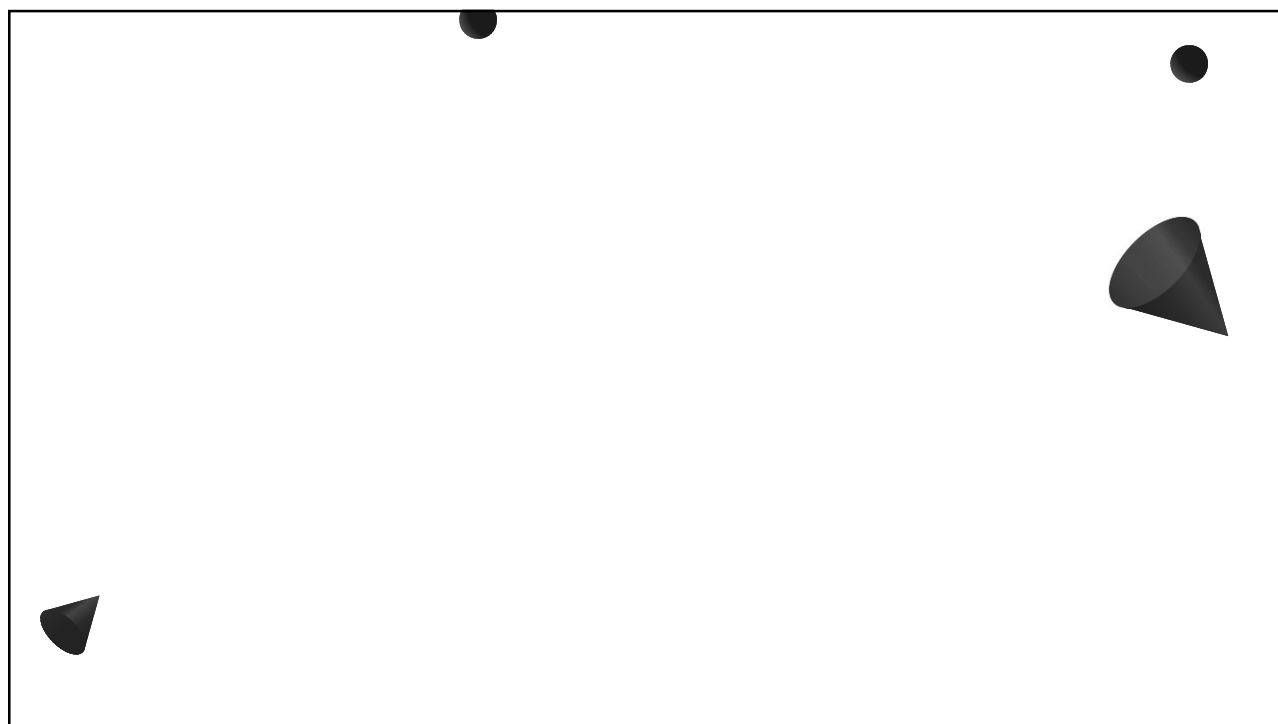


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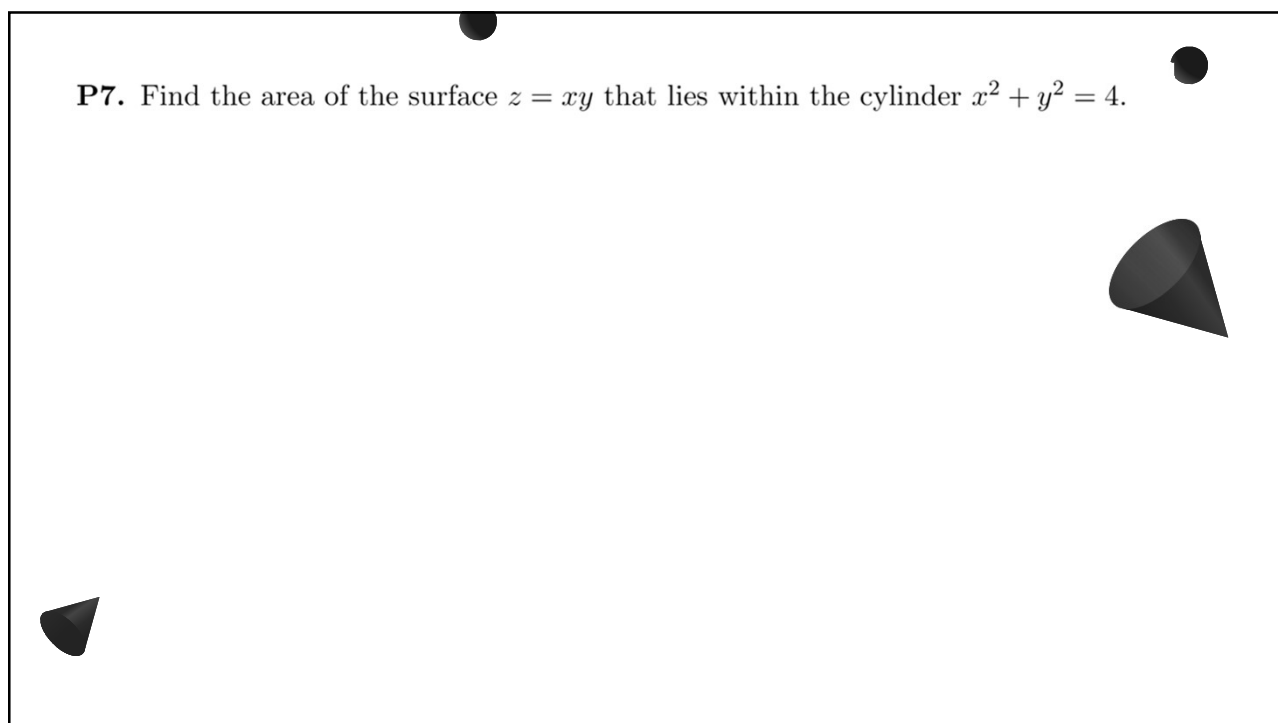
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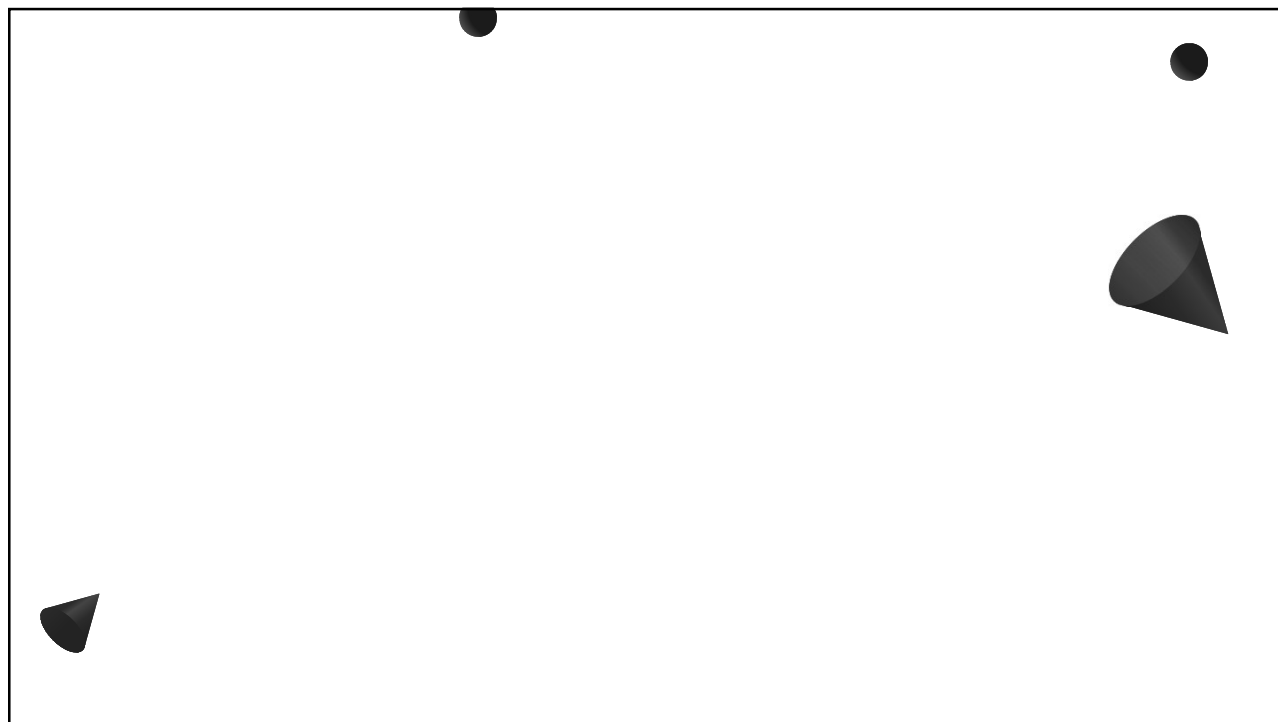
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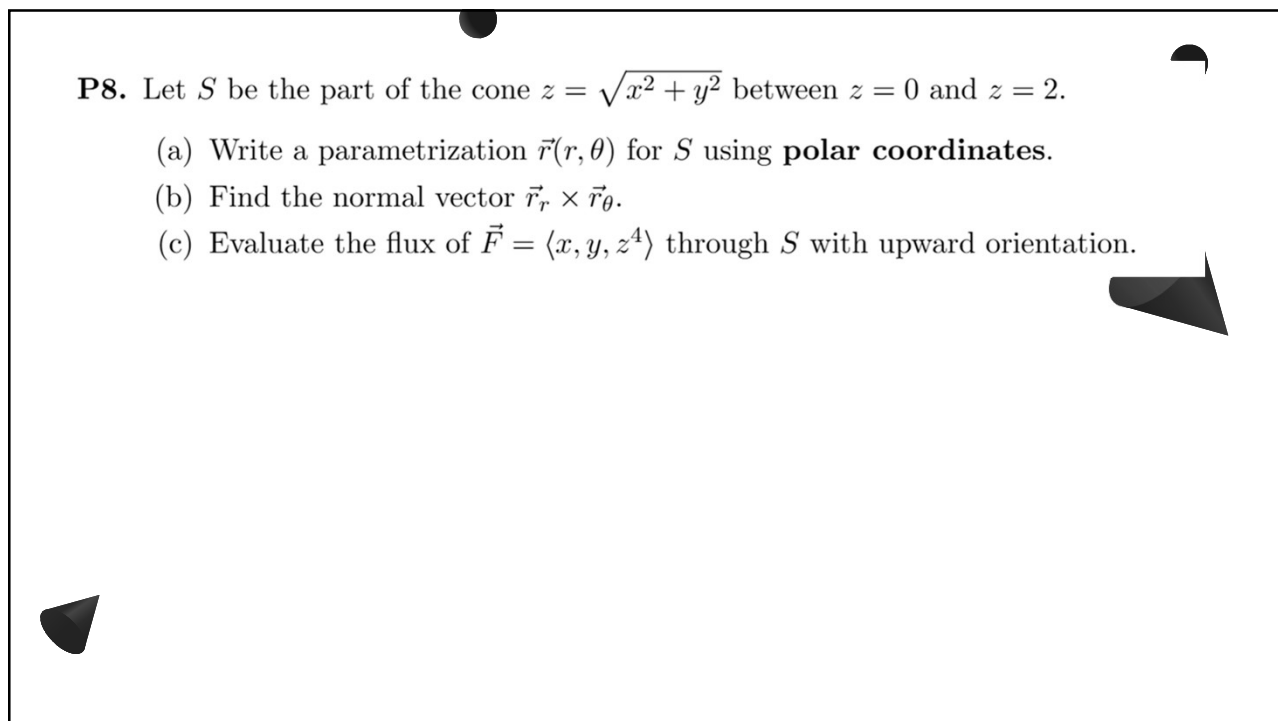
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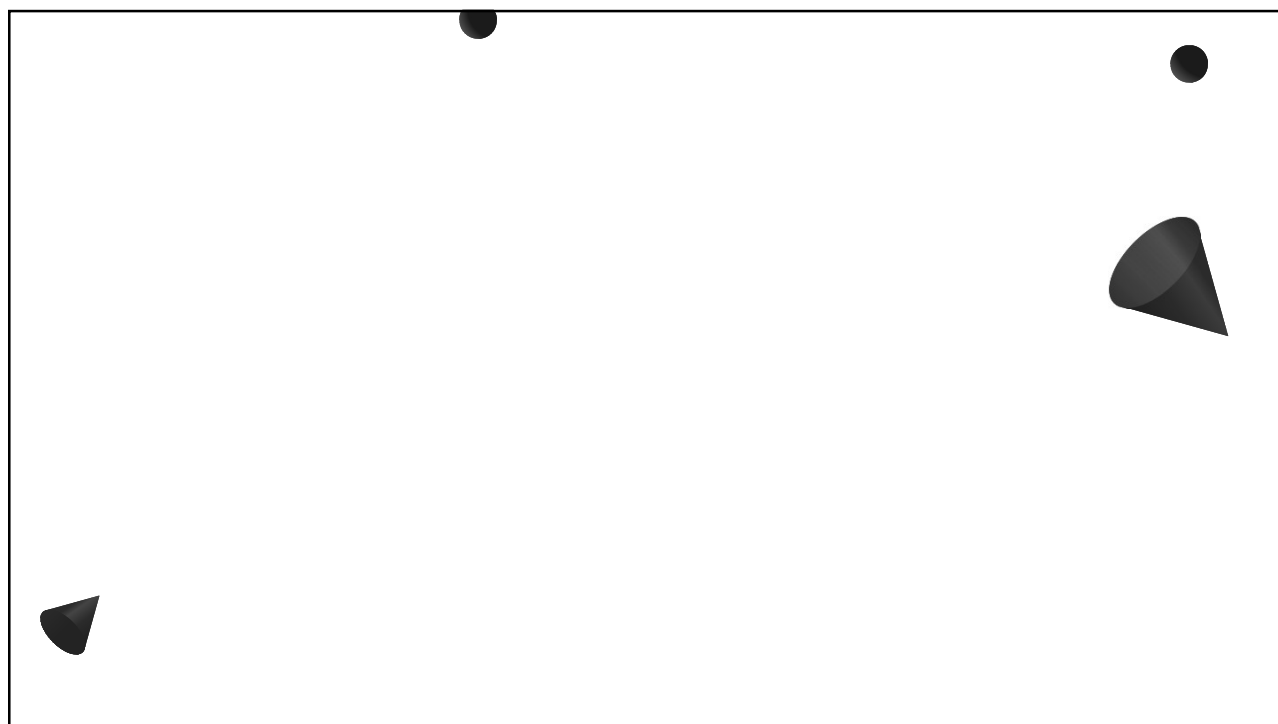
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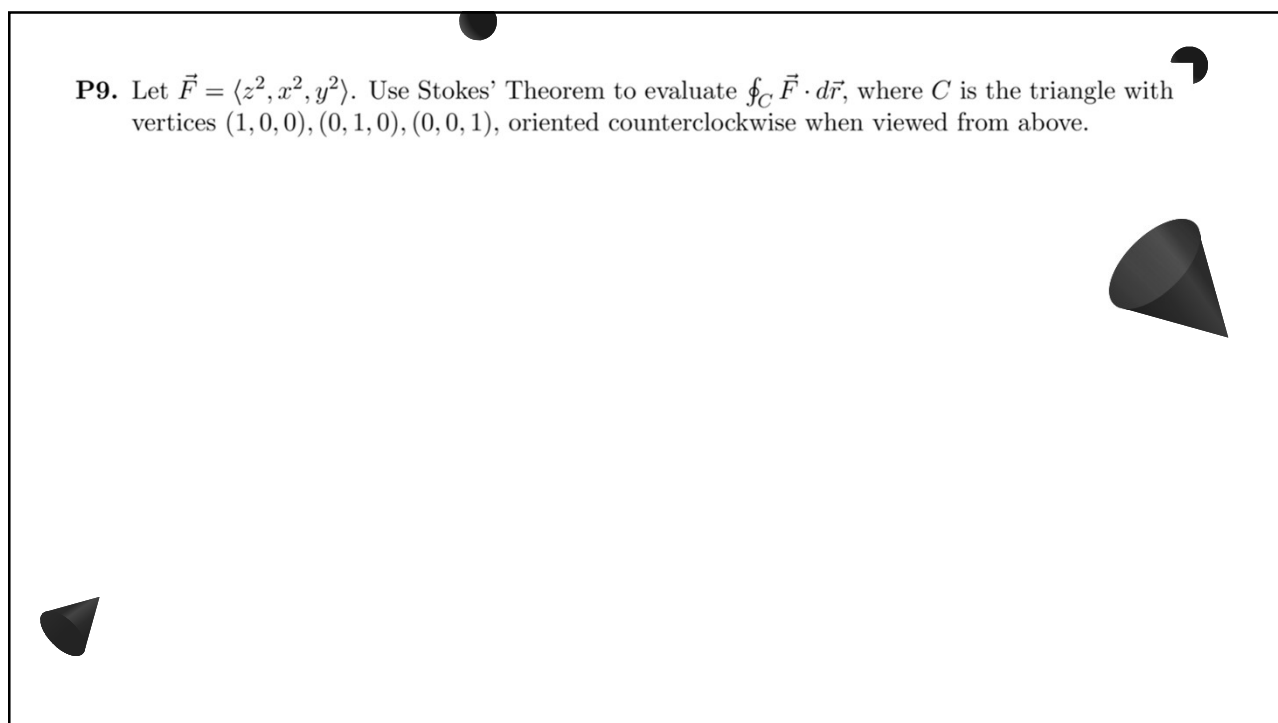
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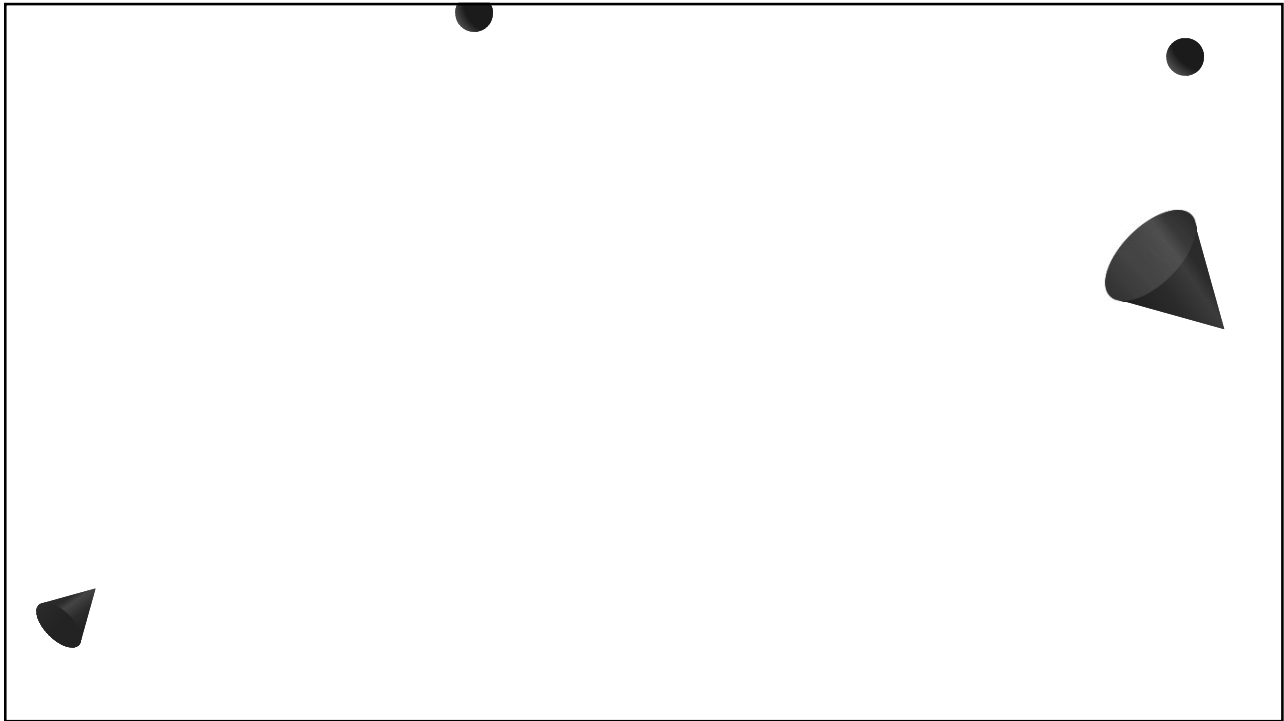
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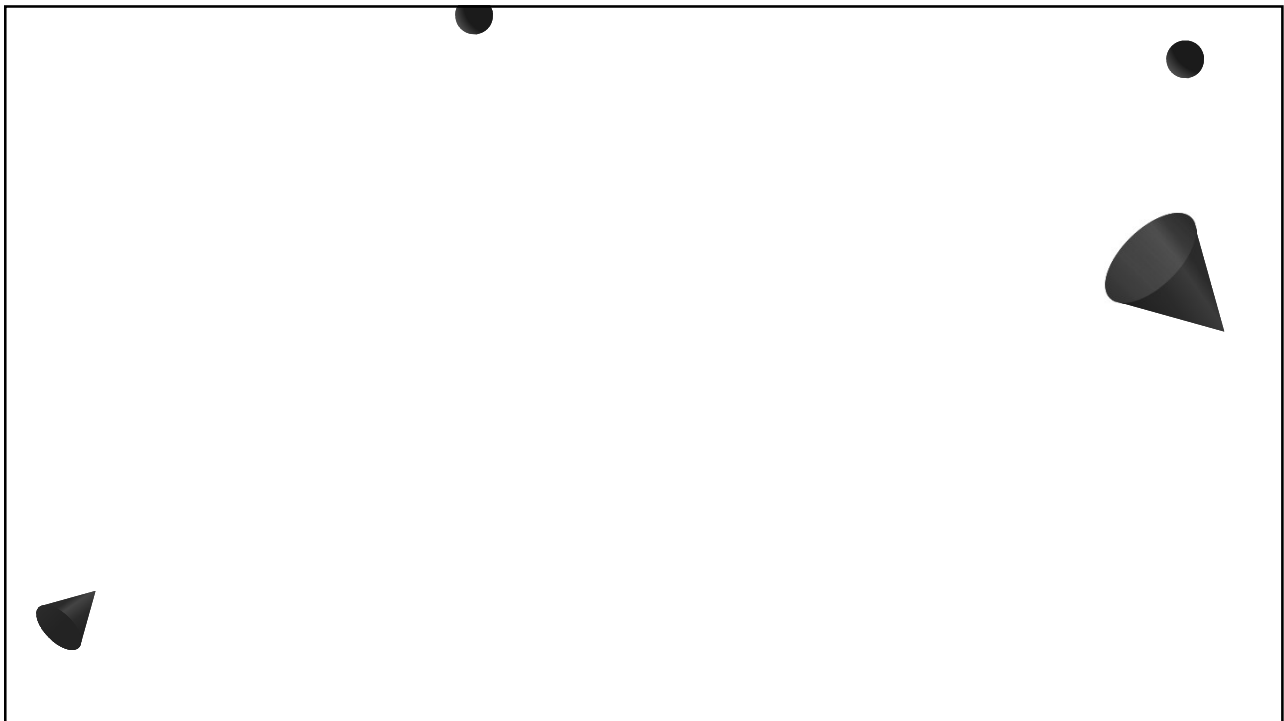
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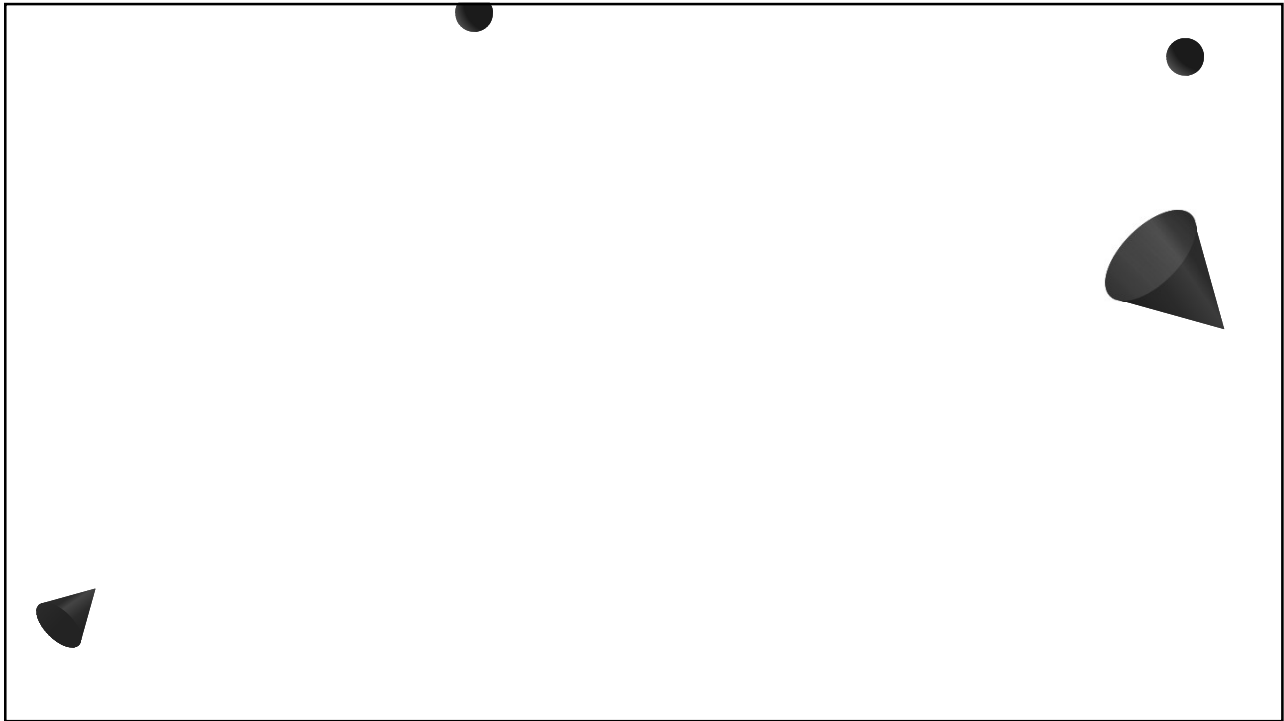
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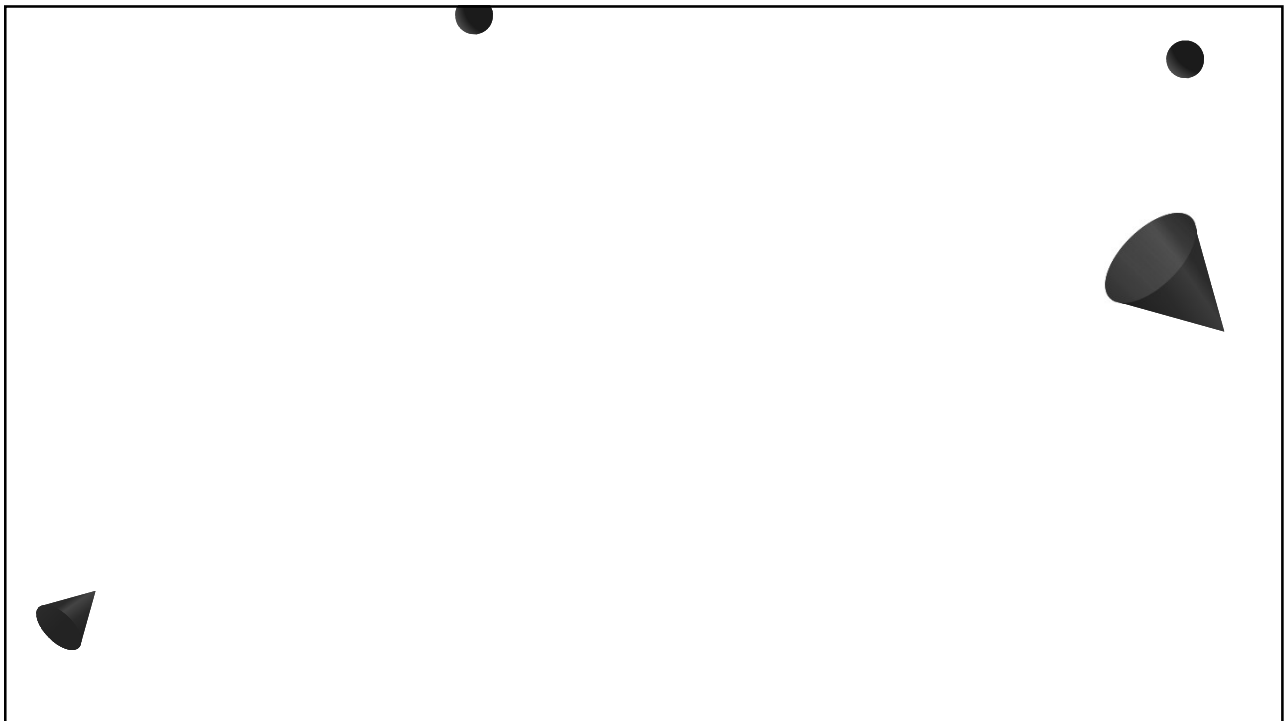
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